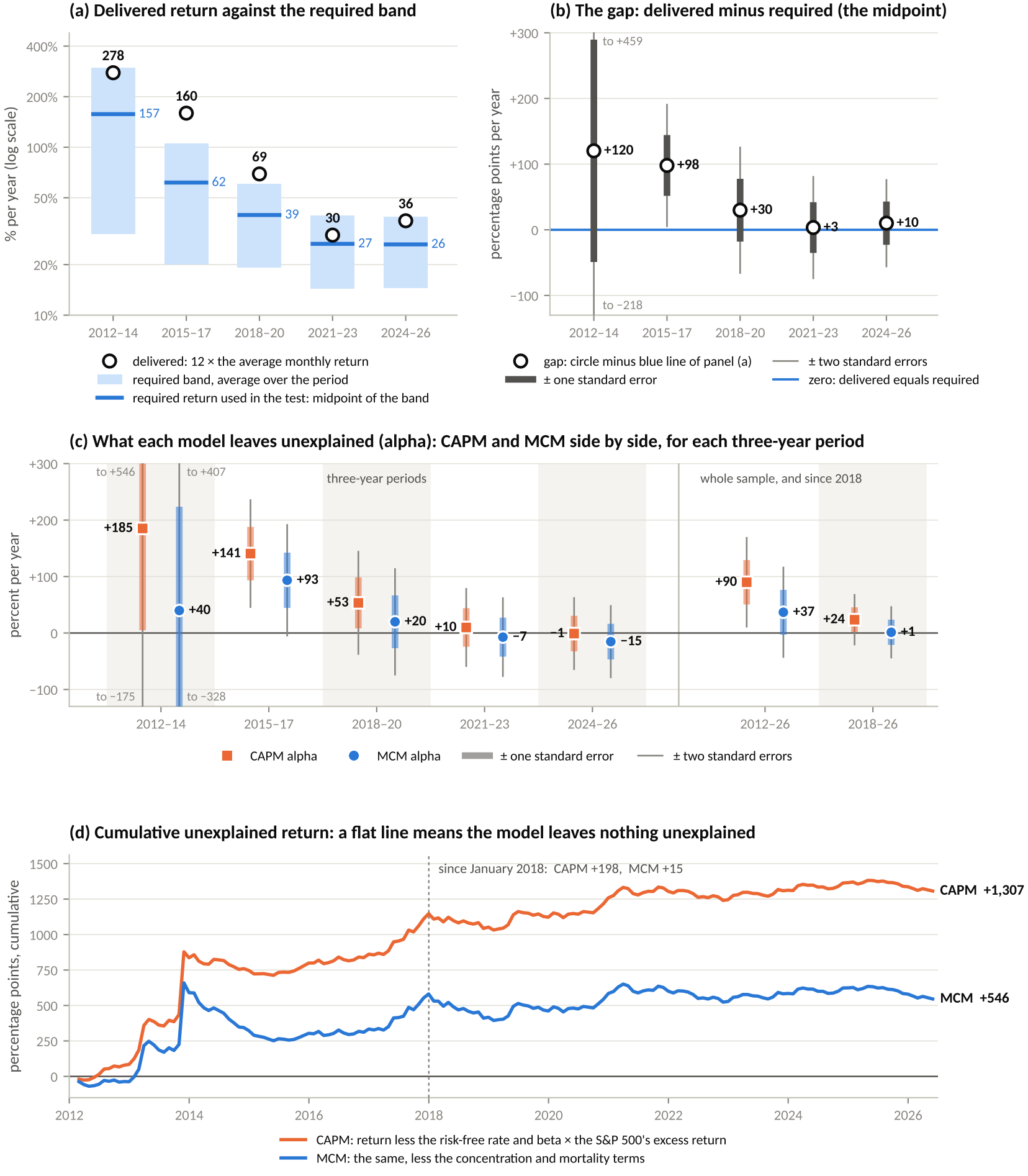

(a) Delivered return against the required band
% per year (log scale)
400%
200%
100%
50%
20%
10%
278
160
69
30
36
157
62
39
27
26
2012–14
2015–17
2018–20
2021–23
2024–26
delivered: 12 × the average monthly return
required band, average over the period
required return used in the test: midpoint of the band
(b) The gap: delivered minus required (the midpoint)
percentage points per year
+300
+200
+100
0
−100
to +459
to −218
+120
+98
+30
+3
+10
2012–14
2015–17
2018–20
2021–23
2024–26
gap: circle minus blue line of panel (a)
± one standard error
± two standard errors
zero: delivered equals required
(c) What each model leaves unexplained (alpha): CAPM and MCM side by side, for each three-year period
percent per year
+300
+200
+100
0
−100
to +546
to +407
to −175
to −328
three-year periods
whole sample, and since 2018
+185
+40
+141
+93
+53
+20
+10
−7
−1
−15
+90
+37
+24
+1
2012–14
2015–17
2018–20
2021–23
2024–26
2012–26
2018–26
CAPM alpha
MCM alpha
± one standard error
± two standard errors
(d) Cumulative unexplained return: a flat line means the model leaves nothing unexplained
percentage points, cumulative
1500
1250
1000
750
500
250
0
since January 2018: CAPM +198, MCM +15
CAPM +1,307
MCM +546
2012
2014
2016
2018
2020
2022
2024
2026
CAPM: return less the risk-free rate and beta × the S&P 500's excess return
MCM: the same, less the concentration and mortality terms